MATH 128a: Homework #5

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Due by: **Wednesday, October 8, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Chapter 1

(a) Show that the sequence $a_n = e^{-\frac{1}{n^2}}$ satisfies $a_n = 1 + O(n^{-2})$ as $n \to \infty$.

(b) Show that the equation $0 = x^2 - 4x^3 + 1$ has at least one solution in the interval $[0, 1]$.

(c) Find the extrema of the function $f(x) = x/(1 + x)$ on the interval $[0, 4]$.

(d) Do problem 10(8) in Section 1.1 of the textbook.

(e) Do problem 18 of Section 1.1 of the textbook.

(f) Do problem 1 b) and c) of Section 1.2 of the textbook.

(g) Solve problem 20b) (16b) of Section 1.2 of the textbook.

Problem 2: Bisection method

(a) What is going wrong in the following code for the bisection method? (aka time to debug!)

Listing 1: Code for bisection method

```
function [p,nit] = bisection(f,a,b,tol)

if nargin < 4
    tol = 10^(-6);
    if nargin < 2
        a=0;
        b=1;
        if nargin < 1
            f = @(x)(x^2-0.5);
        end
    end
end
```

```
% solve f(p)=0 using the bisection method, stop when we find a point that
% is closer than tol to p
% nit number of iterations needed


for k = 1:10000
    p = (a+b)/2;
    nit = k;
    if abs(p-a)< tol || f(p)==0, break; end
    if f(a)*f(b)>0
        a = p;
    else
        b = p;
    end
end
```

(b) Implement a different stopping criterion than the one used in this code and compare to the original one.

Problem 3: Fixed point iteration

Solve problem 3 of Section 2.2 of the textbook.

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